

Scenario 1: Student Academic Performance Dashboard

Learning Outcome

Understand how position and color aesthetics represent different data types.

Scenario

The Head of the Computer Science Department wants to analyze whether students with higher attendance perform better in examinations. The department also wants to compare the performance across different programs.

Dataset

- Student ID
- Attendance (%)
- Internal Marks
- Department (CSE, AI, IT, ECE)

Student Task

- Identify quantitative and categorical variables.
- Map Attendance to the X-axis.
- Map Marks to the Y-axis.
- Represent the Department using different colors.
- Add an appropriate title, labels, and legend.

Attendance vs internal marks by department

R code:

```
library(ggplot2)
```

```
set.seed(42)
```

```
n <- 100
```

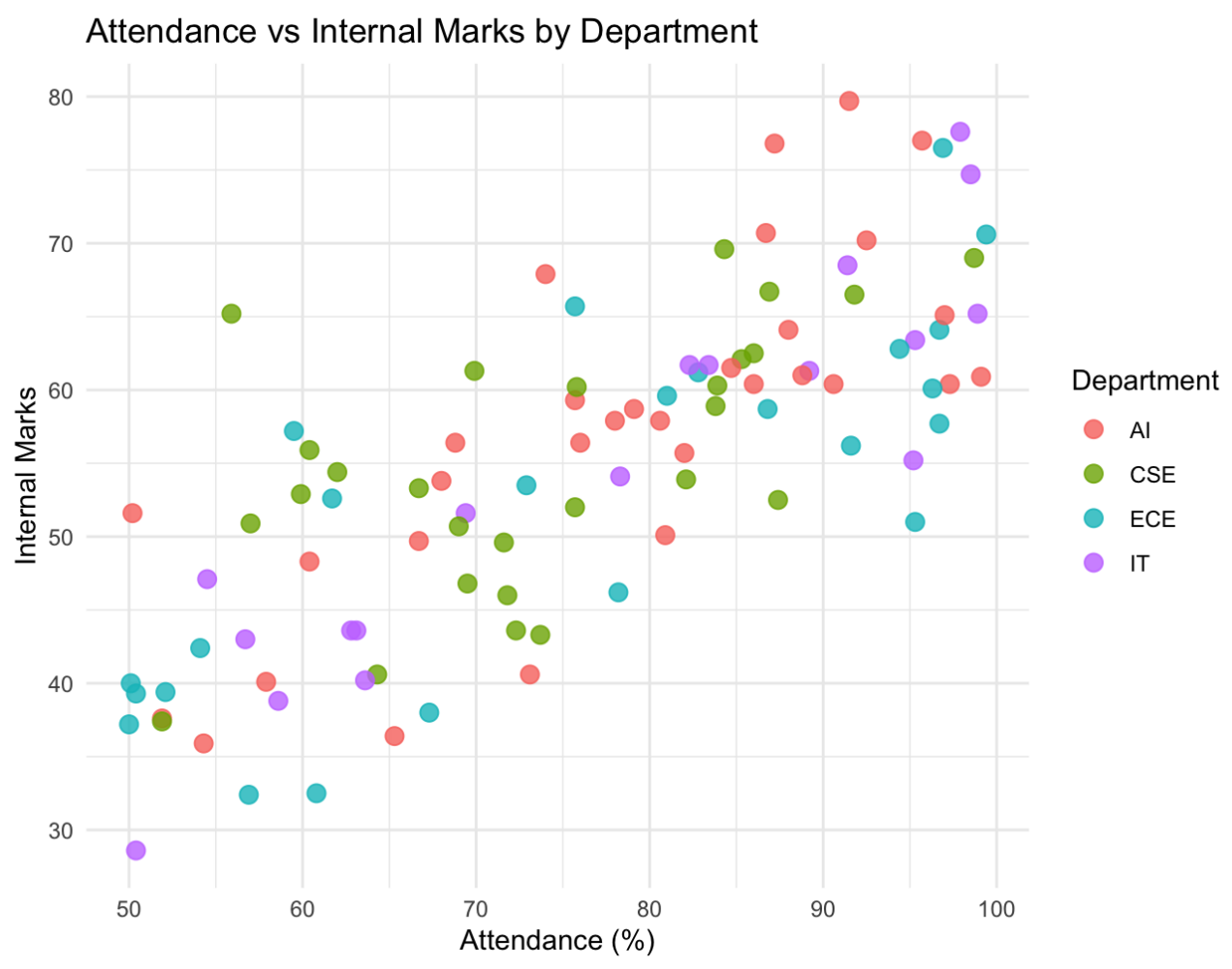
```
student_data <- data.frame(  
  Student_ID = paste0("S", 1:n),  
  Attendance = round(runif(n, 50, 100), 1),  
  Department = sample(c("CSE", "AI", "IT", "ECE"), n, replace = TRUE)  
)
```

```
student_data$Internal_Marks <- round(student_data$Attendance * 0.6 + rnorm(n, 10, 8), 1)  
student_data$Internal_Marks <- pmin(pmax(student_data$Internal_Marks, 0), 100)
```

```
str(student_data)
```

```
ggplot(student_data, aes(x = Attendance, y = Internal_Marks, color = Department)) +  
  geom_point(size = 3, alpha = 0.8) +  
  labs(  
    title = "Attendance vs Internal Marks by Department",  
    x = "Attendance (%)",  
    y = "Internal Marks",  
    color = "Department"  
  ) +  
  theme_minimal()
```

output:



Scenario 2: Electric Vehicle Market Analysis

Learning Outcome

Learn how size and color aesthetics can simultaneously represent multiple variables.

Scenario

An automobile company is studying the relationship between battery capacity and vehicle price

to understand whether larger batteries significantly increase the selling price.

Dataset

- Vehicle Model
- Battery Capacity (kWh)
- Price
- Driving Range
- Manufacturer

Student Task

- Plot Battery Capacity against Price.
- Use Driving Range as bubble size.
- Use Manufacturer as color.
- Interpret whether larger batteries correspond to higher prices.

code:

```
library(ggplot2)
```

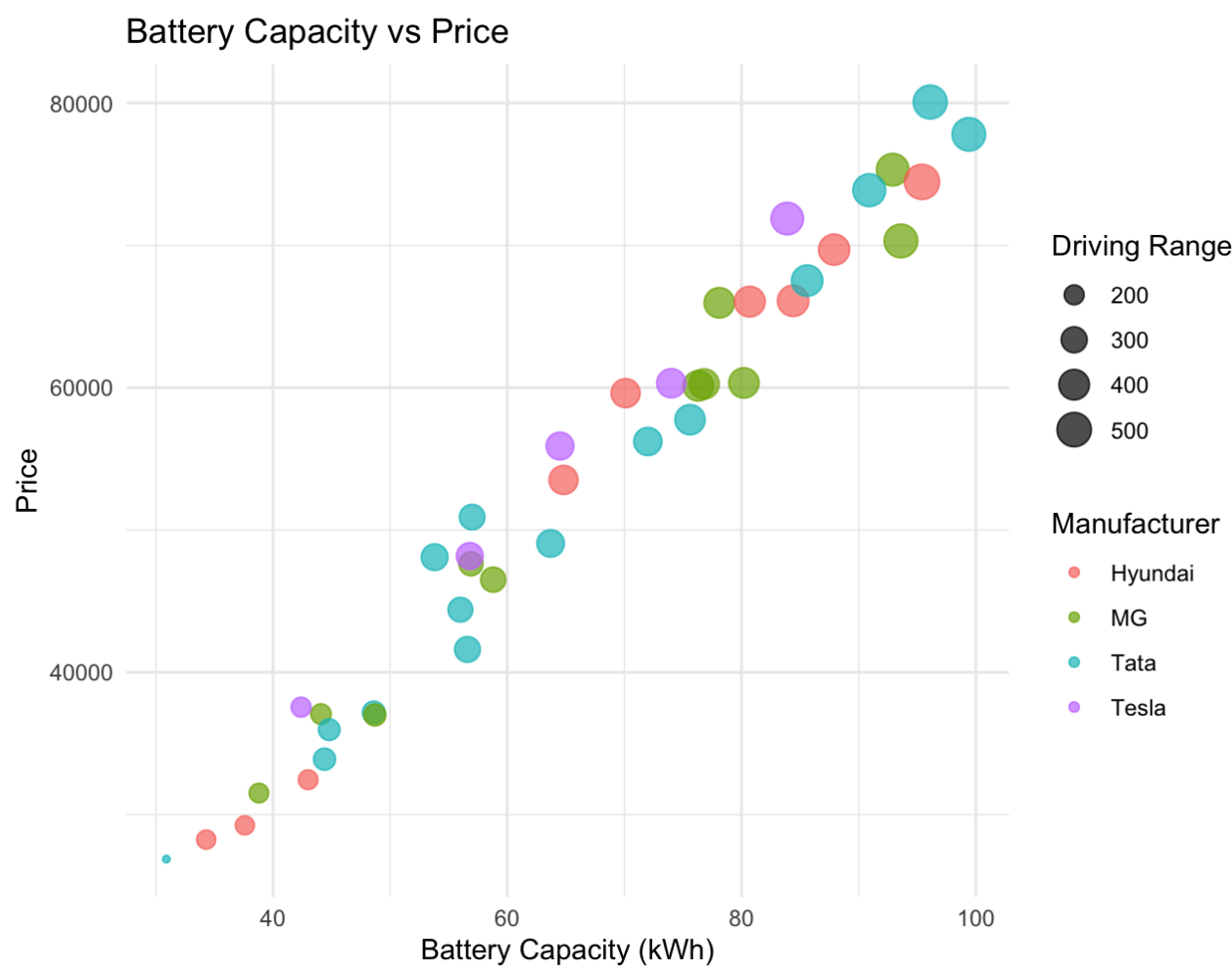
```
set.seed(1)
```

```
n <- 40
```

```
ev_data <- data.frame(  
  Vehicle_Model = paste0("EV", 1:n),  
  Battery_Capacity = round(runif(n, 30, 100), 1),  
  Manufacturer = sample(c("Tesla", "Tata", "Hyundai", "MG"), n, replace = TRUE)  
)  
ev_data$Driving_Range <- round(ev_data$Battery_Capacity * 5 + rnorm(n, 0, 20),  
0)  
ev_data$Price <- round(ev_data$Battery_Capacity * 800 + rnorm(n, 0, 3000), 0)
```

```
ggplot(ev_data, aes(x = Battery_Capacity, y = Price, size = Driving_Range, color =  
Manufacturer)) +  
  geom_point(alpha = 0.7) +  
  labs(title = "Battery Capacity vs Price", x = "Battery Capacity (kWh)", y = "Price",  
        size = "Driving Range", color = "Manufacturer") +  
  theme_minimal()
```

output:



Scenario 3: Hospital Patient Admission Analysis

Learning Outcome

Visualize categorical variables using fill color and position aesthetics.

Scenario

A hospital administrator wants to compare patient admissions across departments during the last month.

Dataset

- Department
- Number of Patients
- Average Treatment Cost

Student Task

- Create a bar chart.
- Use different fill colors for each department.
- Display treatment cost as labels on the bars.

• code:

```
library(ggplot2)
```

```
hospital_data <- data.frame(
```

```
  Department = c("Cardiology", "Neurology", "Orthopedics", "Pediatrics", "General"),
```

```
  Number_of_Patients = c(120, 85, 100, 150, 200),
```

```
  Average_Treatment_Cost = c(45000, 60000, 35000, 20000, 15000)
```

```
)
```

```
ggplot(hospital_data, aes(x = Department, y = Number_of_Patients, fill = Department)) +  
  geom_col() +
```

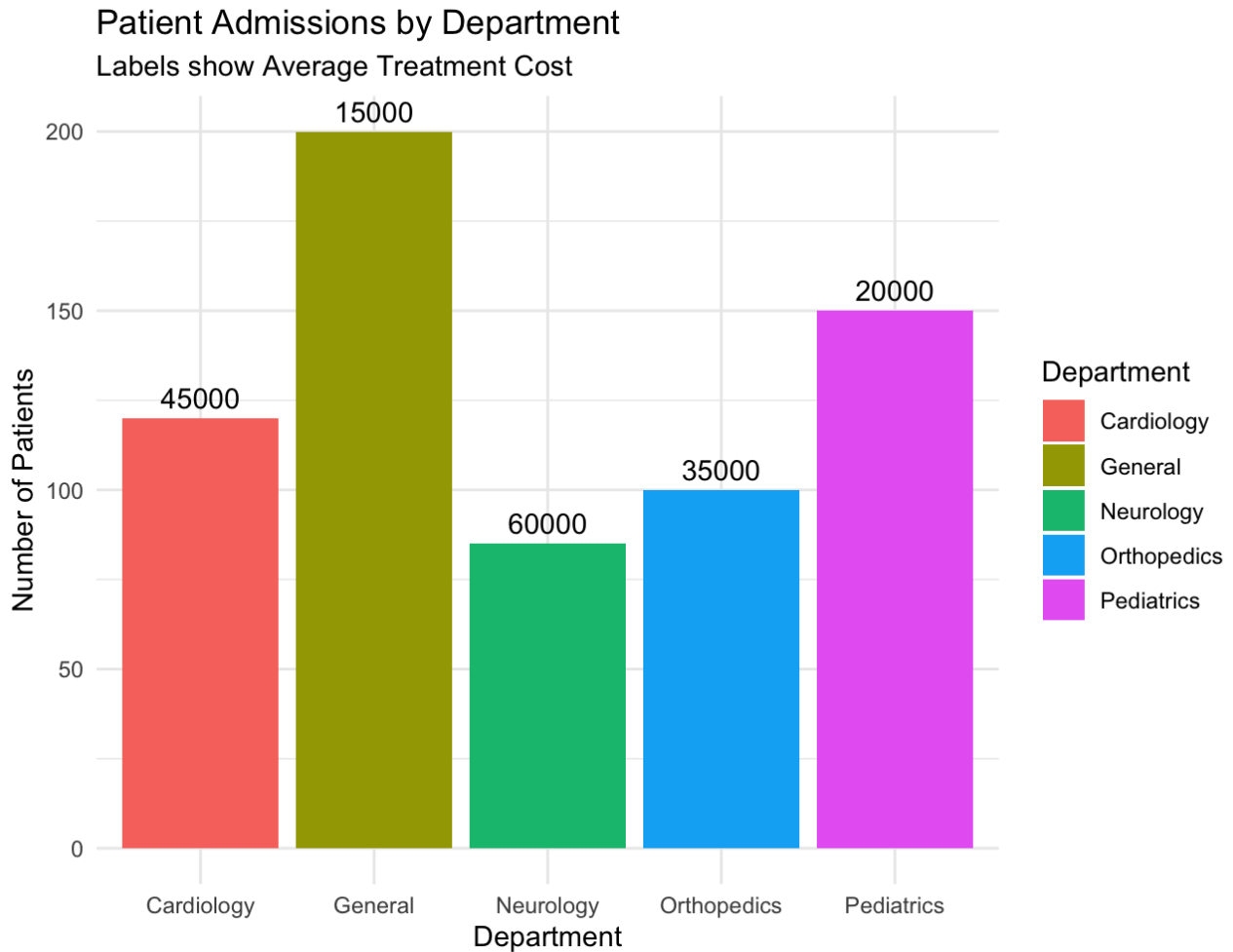
```
  geom_text(aes(label = Average_Treatment_Cost), vjust = -0.5) +
```

```
  labs(title = "Patient Admissions by Department", subtitle = "Labels show Average Treatment  
Cost",
```

```
    x = "Department", y = "Number of Patients") +
```

```
  theme_minimal()
```

output:



Scenario 4: Weather Monitoring System

Learning Outcome

Understand how line type and color distinguish multiple categories.

Scenario

The Indian Meteorological Department has collected daily temperature data for three cities over one month and wants to compare climate trends.

Dataset

- Date
- Temperature
- City

Student Task

- Draw a line chart.
- Use different colors for each city.
- Use different line styles for each city.
- Compare temperature trends.

Code:

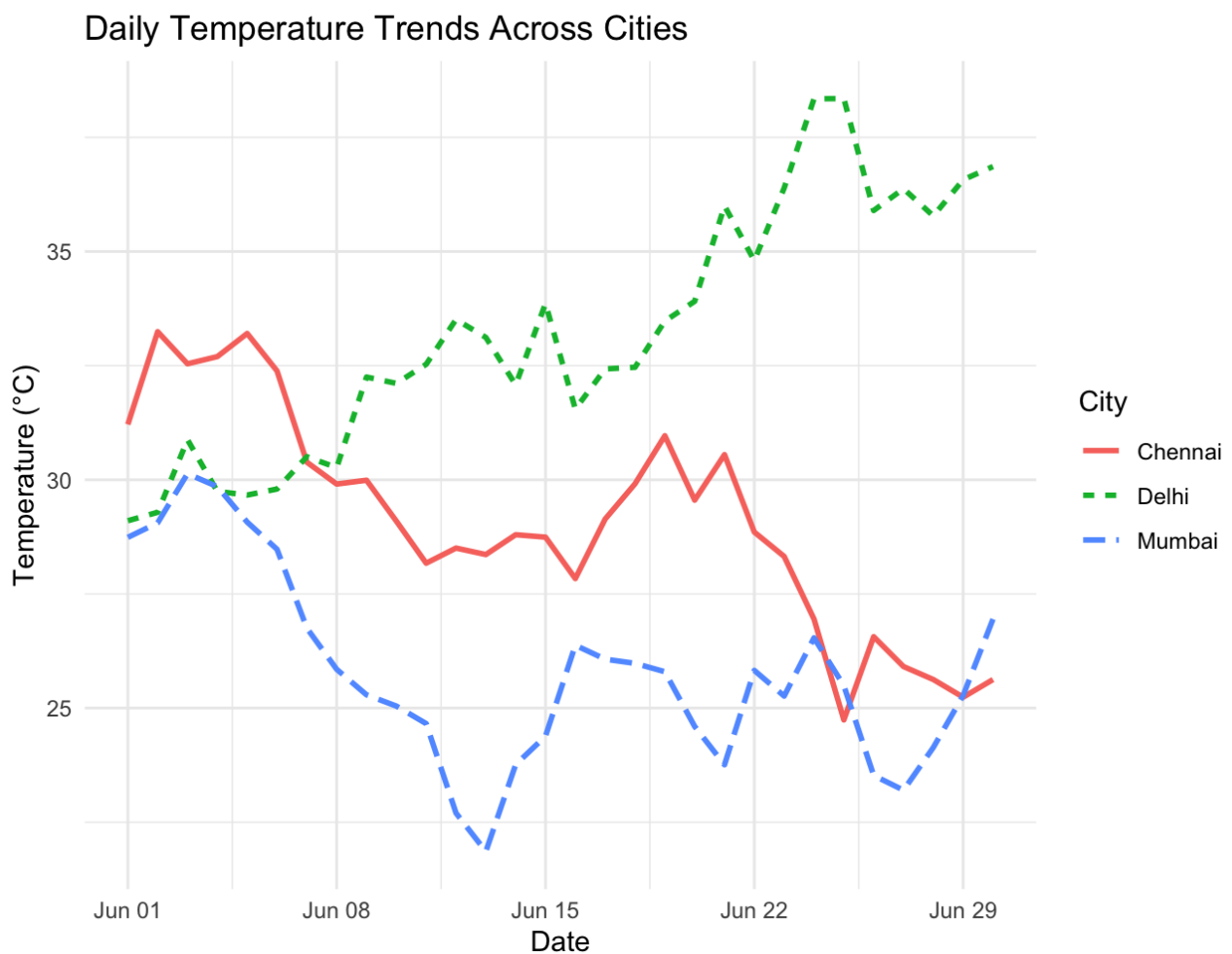
```
library(ggplot2)

set.seed(2)
dates <- rep(seq(as.Date("2026-06-01"), by = "day", length.out = 30), 3)
cities <- rep(c("Delhi", "Mumbai", "Chennai"), each = 30)
temperature <- c(
  30 + cumsum(rnorm(30, 0, 1)),
  28 + cumsum(rnorm(30, 0, 1)),
  33 + cumsum(rnorm(30, 0, 1))
)

weather_data <- data.frame(Date = dates, City = cities, Temperature = temperature)

ggplot(weather_data, aes(x = Date, y = Temperature, color = City, linetype = City)) +
  geom_line(linewidth = 1) +
  labs(title = "Daily Temperature Trends Across Cities", x = "Date", y = "Temperature (°C)") +
  theme_minimal()
```

output:



Scenario 5: Supermarket Sales Analysis

Learning Outcome

Apply position and fill aesthetics for grouped comparisons.

Scenario

A supermarket chain wants to identify which product categories generate the highest monthly

sales across different branches.

Dataset

- Month
- Product Category
- Sales Amount
- Branch

Student Task

- Create grouped bar charts.
- Use branch as fill color.
- Compare product sales among branches.

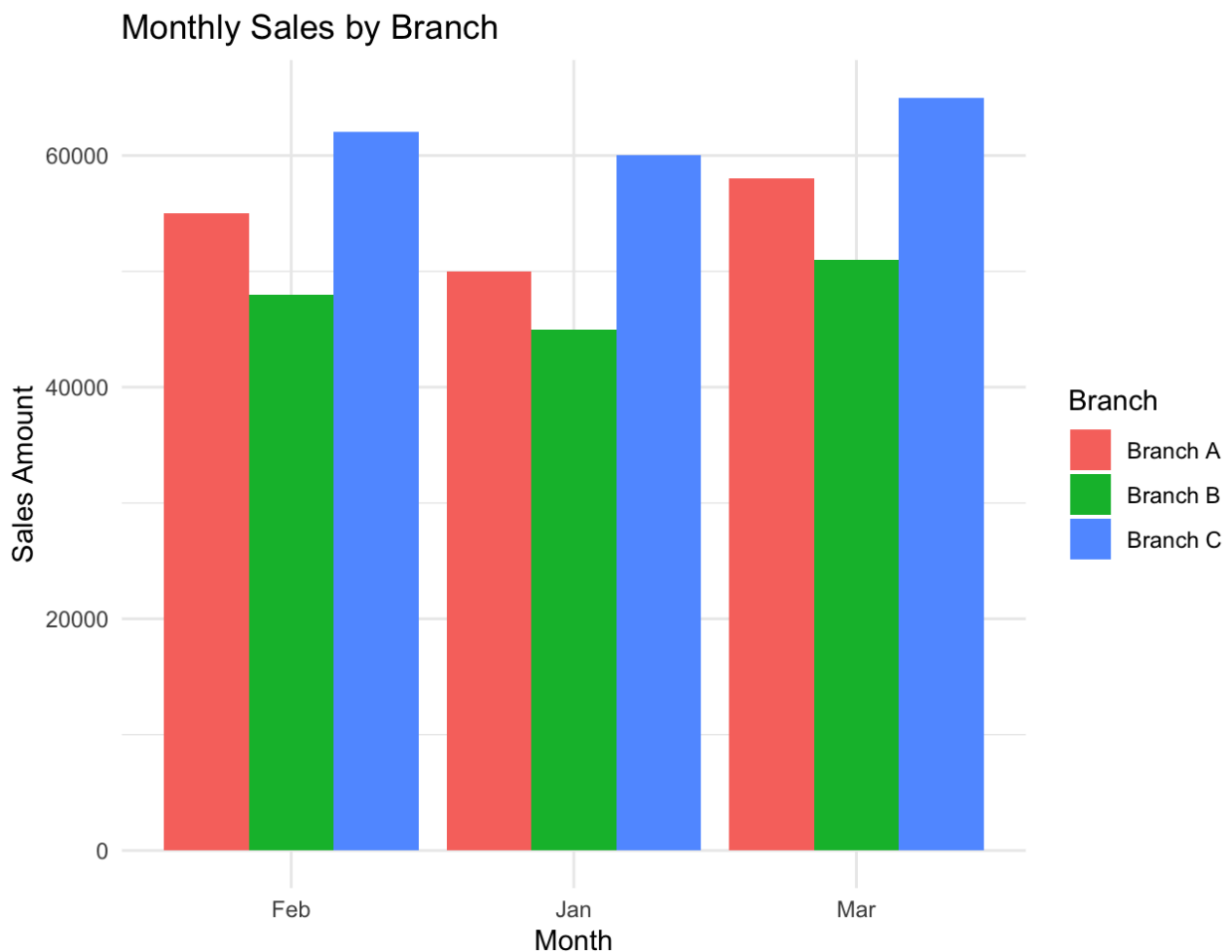
code:

```
library(ggplot2)
```

```
sales_data <- data.frame(  
  Month = rep(c("Jan", "Feb", "Mar"), each = 3),  
  Branch = rep(c("Branch A", "Branch B", "Branch C"), times = 3),  
  Sales_Amount = c(50000, 45000, 60000, 55000, 48000, 62000, 58000, 51000, 65000)  
)
```

```
ggplot(sales_data, aes(x = Month, y = Sales_Amount, fill = Branch)) +  
  geom_col(position = "dodge") +  
  labs(title = "Monthly Sales by Branch", x = "Month", y = "Sales Amount") +  
  theme_minimal()
```


output:



Scenario 6: Employee Salary Distribution

Learning Outcome

Understand continuous scales and histogram binning.

Scenario

The HR department wants to understand salary distribution among employees to determine whether salaries are evenly distributed across the organization.

Dataset

- Employee ID
- Salary
- Experience
- Department

Student Task

- Create a histogram of salaries.
- Experiment with different bin sizes.
- Compare the effect of changing the scale.

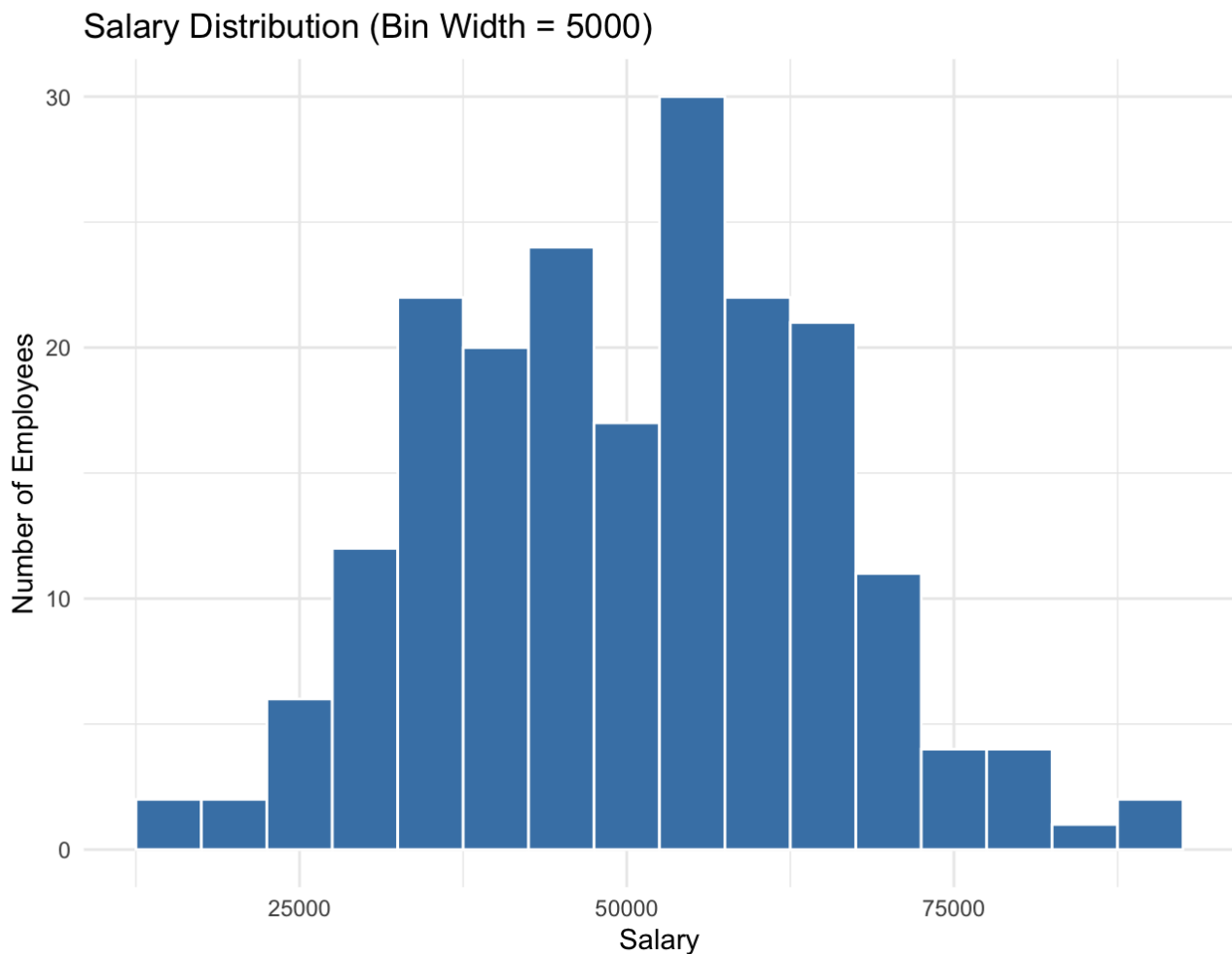
Code:

```
library(ggplot2)

set.seed(3)
n <- 200
salary_data <- data.frame(
  Employee_ID = 1:n,
  Salary = round(rnorm(n, mean = 50000, sd = 15000)),
  Experience = round(runif(n, 1, 20)),
  Department = sample(c("HR", "IT", "Sales", "Finance"), n, replace = TRUE)
)

# Try changing binwidth: 2000, 5000, 10000
ggplot(salary_data, aes(x = Salary)) +
  geom_histogram(binwidth = 5000, fill = "steelblue", color = "white") +
  labs(title = "Salary Distribution (Bin Width = 5000)", x = "Salary", y = "Number of Employees") +
  theme_minimal()
```

output:



Scenario 7: Smartphone Market Comparison

Learning Outcome

Map multiple variables using shape, size, and color.

Scenario

A technology company wants to compare smartphone specifications to understand how RAM and processor speed influence product pricing.

Dataset

- Brand
- RAM
- Processor Speed
- Price
- Storage

Student Task

- Plot RAM against Price.
- Represent Brand using shapes.
- Represent Storage using point size.
- Use color to indicate Processor Speed.

code:

```
library(ggplot2)
```

```
set.seed(4)
```

```
n <- 50
```

```
phone_data <- data.frame(
```

```
  Brand = sample(c("Samsung", "Apple", "OnePlus", "Xiaomi"), n, replace = TRUE),
```

```
  RAM = sample(c(4, 6, 8, 12, 16), n, replace = TRUE),
```

```
  Storage = sample(c(64, 128, 256, 512), n, replace = TRUE),
```

```
  Processor_Speed = round(runif(n, 1.8, 3.2), 1)
```

```
)
```

```
phone_data$Price <- round(phone_data$RAM * 3000 + phone_data$Storage * 20 + rnorm(n, 0, 2000))
```

```
ggplot(phone_data, aes(x = RAM, y = Price, shape = Brand, size = Storage, color = Processor_Speed)) +
```

```
  geom_point(alpha = 0.8) +
```

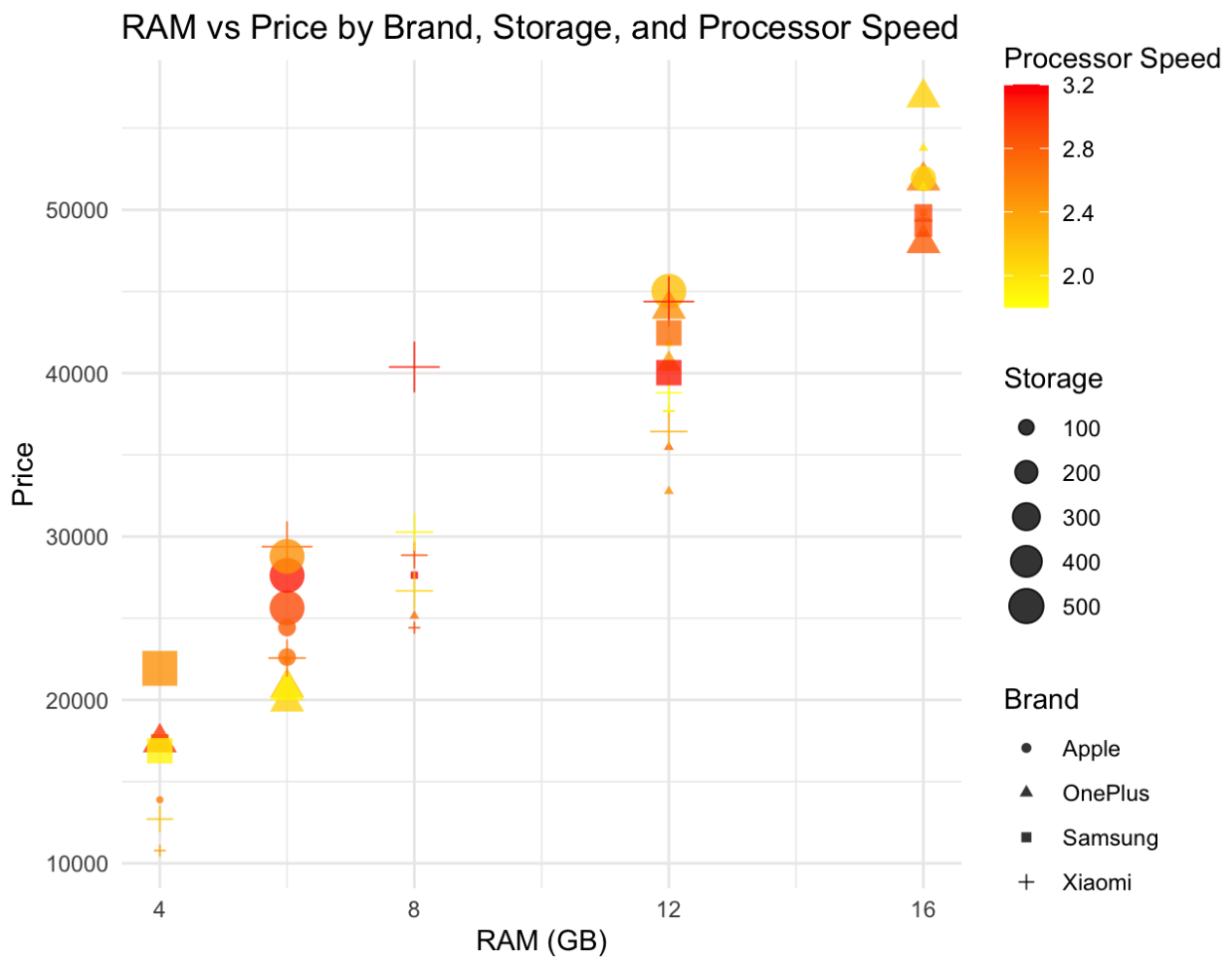
```
  scale_color_gradient(low = "yellow", high = "red") +
```

```
  labs(title = "RAM vs Price by Brand, Storage, and Processor Speed",
```

```
        x = "RAM (GB)", y = "Price", color = "Processor Speed") +
```

```
  theme_minimal()
```

Output:



Scenario 8: Online Movie Rating Analysis

Learning Outcome

Understand continuous color scales and gradient aesthetics.

Scenario

A movie streaming platform wants to compare average ratings received by different movie genres over the past year.

Dataset

- Genre
- Average Rating
- Number of Reviews
- Release Year

Student Task

- Create a heatmap.
- Represent ratings using a color gradient.
- Identify the highest-rated genres.

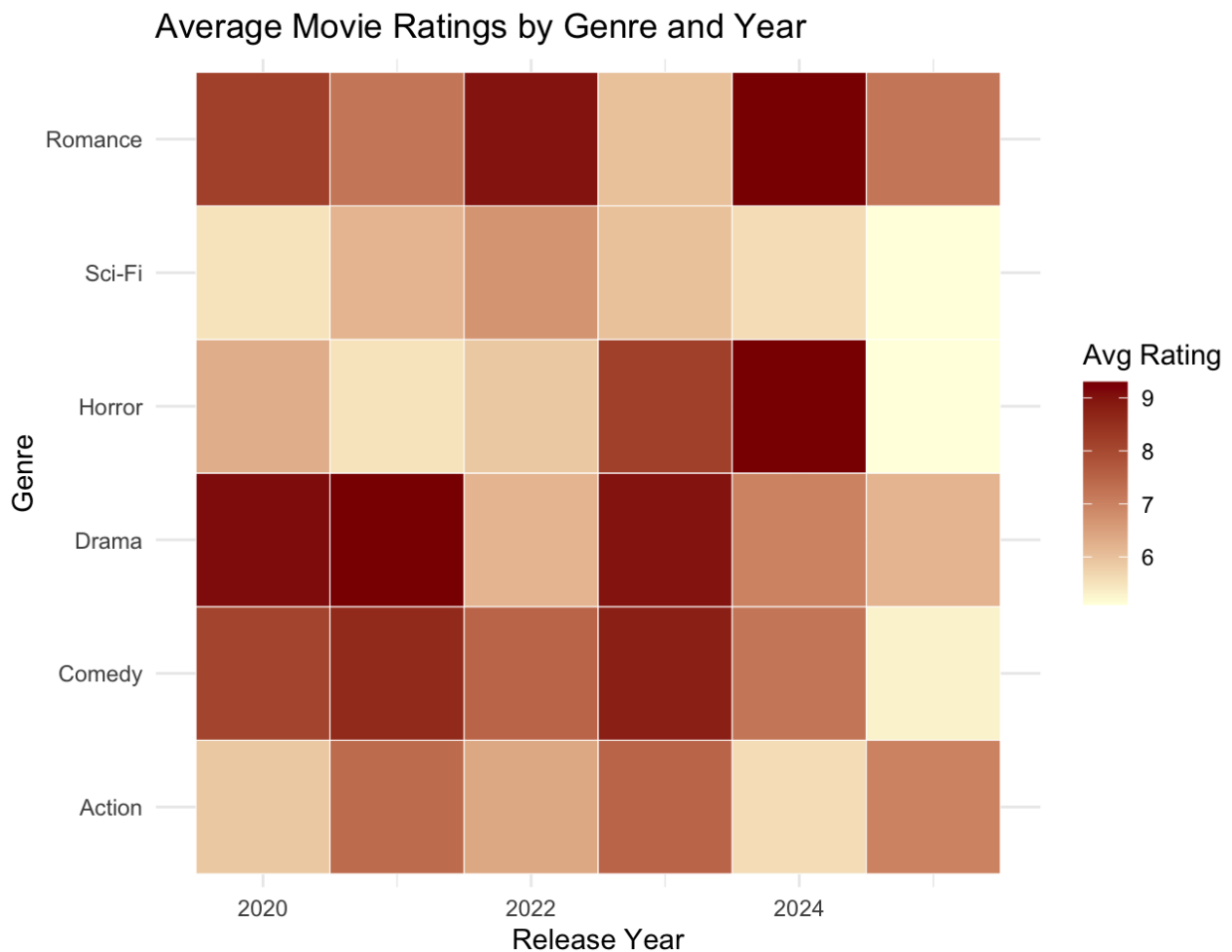
code:

```
library(ggplot2)

set.seed(5)
genres <- c("Action", "Comedy", "Drama", "Horror", "Sci-Fi", "Romance")
years <- 2020:2025
movie_data <- expand.grid(Genre = genres, Release_Year = years)
movie_data$Average_Rating <- round(runif(nrow(movie_data), 5, 9.5), 1)

ggplot(movie_data, aes(x = Release_Year, y = Genre, fill = Average_Rating)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "lightyellow", high = "darkred") +
  labs(title = "Average Movie Ratings by Genre and Year", x = "Release Year", y = "Genre",
       fill = "Avg Rating") +
  theme_minimal()
```

output:



Scenario 9: COVID-19 State-wise Analysis

Learning Outcome

Understand sequential, diverging, and categorical color scales.

Scenario

The Ministry of Health wants to visualize the distribution of COVID-19 cases across different Indian states to identify highly affected regions.

Dataset

- State
- Active Cases
- Vaccination Percentage
- Population

Student Task

- Create a choropleth map (or tile plot if maps are unavailable).
- Represent Active Cases using a sequential color scale.
- Compare the intensity across states.

code:

```
library(ggplot2)
```

```
set.seed(6)
```

```
states <- c("Maharashtra", "Kerala", "Delhi", "Tamil Nadu", "UP", "Karnataka", "West Bengal")
```

```
covid_data <- data.frame(
```

```
  State = states,
```

```
  Active_Cases = round(runif(length(states), 500, 10000)),
```

```
  Vaccination_Percentage = round(runif(length(states), 40, 95), 1)
```

```
)
```

```
ggplot(covid_data, aes(x = State, y = "Cases", fill = Active_Cases)) +
```

```
  geom_tile(color = "white") +
```

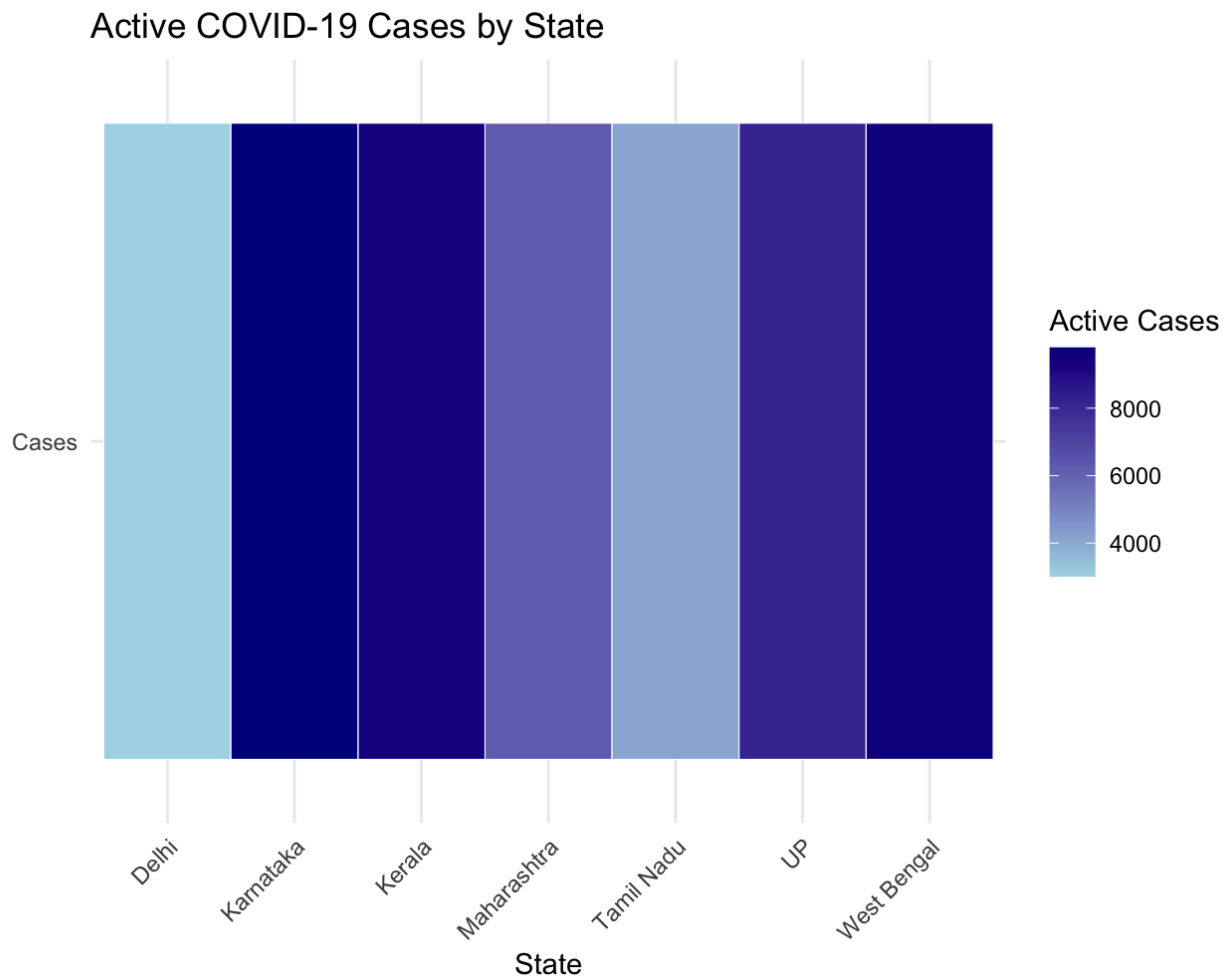
```
  scale_fill_gradient(low = "lightblue", high = "darkblue") +
```

```
  labs(title = "Active COVID-19 Cases by State", x = "State", y = "", fill = "Active Cases") +
```

```
  theme_minimal() +
```

```
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

output:



Scenario 10: Iris Flower Classification

Learning Outcome

Understand how multiple aesthetic mappings improve pattern recognition and class separation.

Scenario

A botanist wants to analyze flower measurements to determine whether different Iris species can be visually separated.

Dataset

(Built-in Iris Dataset)

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width
- Species

Student Task

- Plot Sepal Length against Petal Length.
- Use Species as color.
- Use Species as shape.
- Use Sepal Width as point size.
- Interpret whether species form distinct clusters.

Code:

```
library(ggplot2)

data(iris)

ggplot(iris, aes(x = Sepal.Length, y = Petal.Length,
                 color = Species, shape = Species, size = Sepal.Width)) +
  geom_point(alpha = 0.8) +
  labs(title = "Iris: Sepal Length vs Petal Length", x = "Sepal Length", y = "Petal Length") +
  theme_minimal()
```

output:

